Design of a 2.45GHz RF (Radio Frequency) Receiver in CMOS 45nm process

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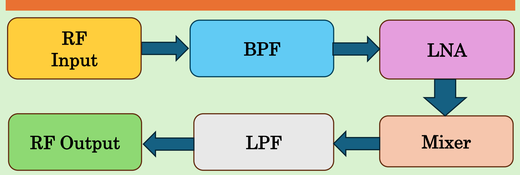
Abstract

The increasing demand for compact and energy-efficient wireless systems requires integrated RF receiver architectures. This work presents a 2.45 GHz CMOS RF receiver implemented in 45nm technology. The architecture integrates amplification, frequency conversion, and filtering stages to process low-power signals. Post-layout simulations in Cadence reveal a voltage gain of 400, a Total Harmonic Distortion (THD) of 1.319%, and a low power consumption of 6.261 mW. These results demonstrate a balanced trade-off between gain and linearity, confirming the design's suitability for wireless applications.

Objective

The objective of this project is to build a 2.45 GHz RF receiver in 45 nm CMOS technology that can detect weak wireless signals and recover useful information from them. The work focuses on improving the performance of key blocks such as the LNA, mixer, and filters to achieve proper amplification, frequency conversion, low noise, and low power consumption.

Methods



- ❑ BPF to isolate the 2.45 GHz spectrum from out-of-band interference.
- ❑ Integrated a LNA to boost the weak signal.
- ❑ Utilized a mixer stage to down convert the modulated signal into I/Q baseband signals.
- ❑ Applied LPF to extract the baseband signals.

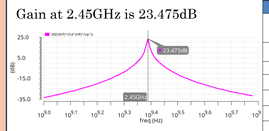
RF Receiver Specification

Frequency	Voltage	Power Consumption	Overall Gain
2.45 GHz	1.2V	<10mW	>100

LNA (Low Noise Amplifier)

AC Response

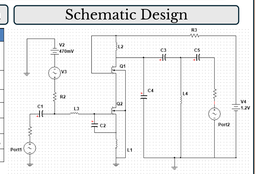
Gain at 2.45GHz is 23.475dB



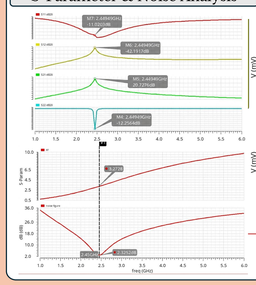
LNA Specification

Parameter	Value
Power Consumption	<2mW
NF (50 Ω)	< 3dB
Gain	15 - 20 dB
S11	≤ -10dB
S22	≤ -10dB
S21	15 – 20 dB
S12	≤ -25dB

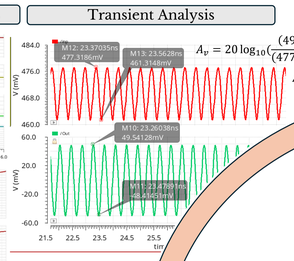
Schematic Design



S-Parameter & Noise Analysis



Transient Analysis

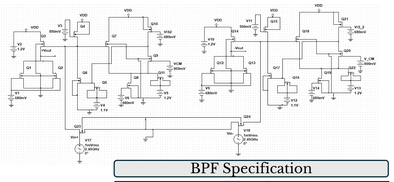


$$A_v = 20\log_{10}\left(\frac{(49.541 + 48.414)mV}{(477.318 - 461.314)mV}\right)$$

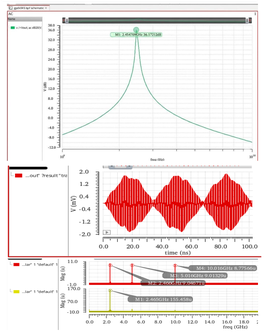
$$A_v = 16.3\ dB$$

BPF (Band Pass Filter)

Schematic Design



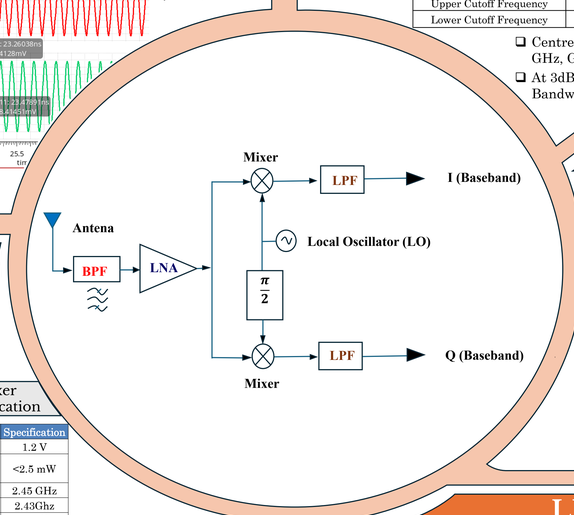
AC Response



BPF Specification

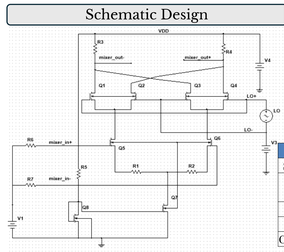
Parameter	Value
Passband Width	20 MHz to 100 MHz
Center Frequency	2.45 GHz
Upper Cutoff Frequency	2.45 GHz + 10 MHz
Lower Cutoff Frequency	2.45 GHz - 10 MHz

- ❑ Centre frequency = 2.45 GHz, Gain = 36.17dB
- ❑ At 3dB Point, Bandwidth = 30 MHz



Gilbert-Cell Mixer

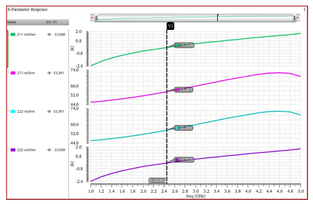
Schematic Design



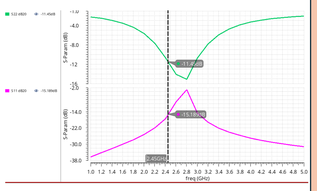
Mixer Specification

Parameter	Specification
Supply voltage	1.2 V
Power Consumption	<2.5 mW
RF frequency	2.45 GHz
Lo Frequency	2.43Ghz
Conversion Gain	>10 dB

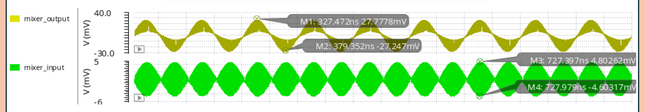
Mixer Input Impedance Matching



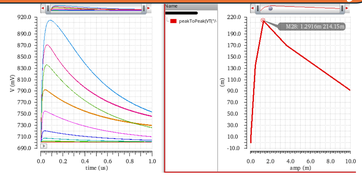
S11 & S22 Parameter



Transient Analysis



Performance Summary



- ❑ Input Range: 1 µm – 1.3 mm

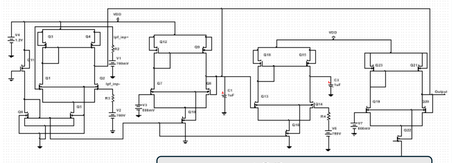
6mW | THD 1.3% | SNR 49dB

LPF (Low Pass Filter)

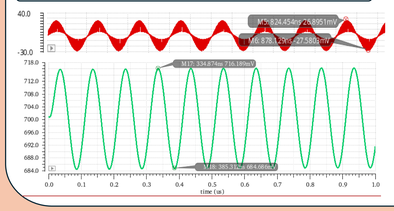
Low Pass Filter (LPF) Specification

Parameter	Specification
Passband Frequency	20 MHz
Stopband Frequency	30 MHz
Stopband Attenuation	> 40 dB
Bandwidth	20 MHz
Passband Ripple	1dB
Source Impedance	50 ohm
Load Impedance	50 ohm

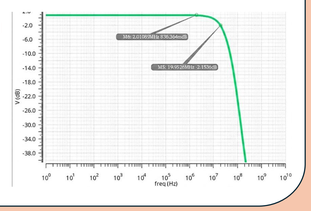
Schematic Design



Transient Analysis

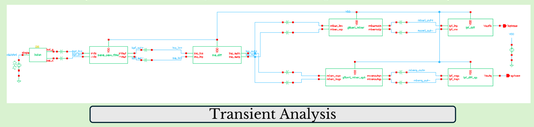


AC Response

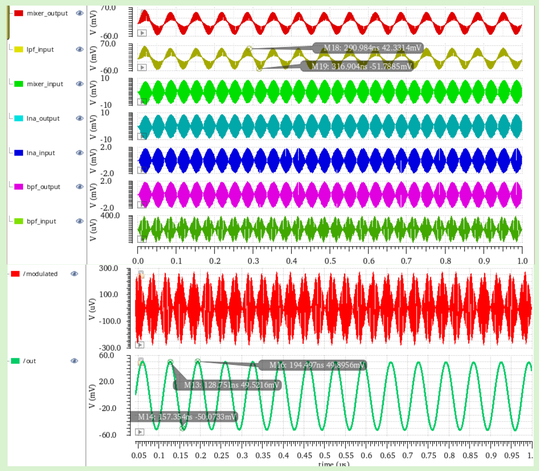


RF Receiver Results

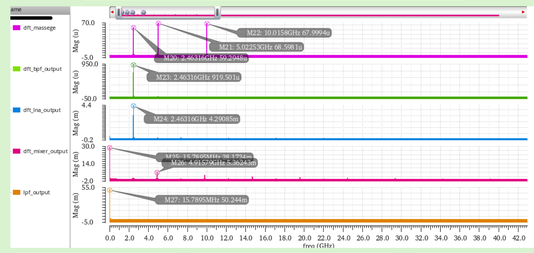
System Block Design



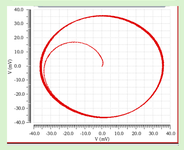
Transient Analysis



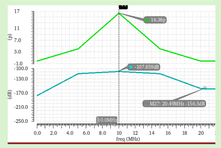
DFT Analysis



Lissajous Figure



PSD Analysis



Total Power

Expression	Value
average(IT("/V35...	6.261E-3

THD

Expression	Value
thd(v("/out" ?res...	1.319

- ➢ The perfect circular Lissajous pattern confirms 0 dB gain mismatch and 0 phase error between the I and Q channels.
- ➢ SNR_dB = P_signal - P_noise = -107.859 -156.5 = 49 dB
- ➢ Power consumption of 6.261 mW and 1.319% THD (Total Harmonic Distortion).

Comparison Table

Category	Parameter	[1]	[2]	[3]	[4]	[5]	[6]	This Work
Process	Technology (nm)	-	180	600	180	180	40	45
	Supply Voltage (V)	-	1.5	3	1.8	1.5	1.1	1.2
System	Frequency (GHz)	2.4	24	2.4	24	2.42	-	2.45
	Total Gain (dB)	35	14.5	12 - 20	25	-	-	400
	Total Power (mW)	40	45	100	25.2	-	-	6.261
LNA	Gain (dB)	-	15	14	14.7	-	-	23.03
	NF (dB)	-	6	3.5	4.3	-	-	2.4
	Topology	-	Common Gate	Common Source Cascode	Transformer-Coupled	-	-	Common Source Inductive Degeneration
Mixer	Conv. Gain (dB)	-	12.5	10	8.7	-	-	
	Type	-	Active Gilbert Cell	Active Gilbert Cell	Current-Bleeding Mixer	-	-	Active Gilbert Cell
BPF	Bandwidth (MHz)	-	-	-	-	38	-	20-100 (Tunable)
	Power Consumption (mW)	-	-	-	-	1.3 (Single Stage)	-	1.63 (Differential)
LPF	Bandwidth (MHz)	-	-	-	-	-	20	20
	Power Consumption (mW)	-	-	-	-	-	2	0.2

Future Direction

Integration of Advanced Blocks
↓
Physical Layout Design
↓
Further Power Optimization

Key Reference

[1] Al-Rawi, Muhanned. "Design of 2.4 GHz Transceiver for Wireless Communication." *Land Forces Academy Review*, vol. 24, no. 3, Nicolae Balcescu Land Forces Academy, 2019, pp. 232-241. https://doi.org/10.2478/raft-2019-0028.

[2] Guan, Xiang, and Ali Hajimiri. "A 24-GHz CMOS Front-End". IEEE Journal of Solid-State Circuits 39 (2): 368–73. https://doi.org/10.1109/JSSC.2003.821783..

[3] Razavi, Behzad. "CMOS RF receiver design for wireless LAN applications." RAWCON 99. IEEE Radio and Wireless Conference (Cat. No. 99EX292).

[4] Chu, Chen-Yuan, et al. "A 24GHz low-power CMOS receiver design "*IEEE International Symposium on Circuits and Systems (ISCAS).*

[5] Bhuiyan, Arif. "Design of a Band-Pass Filter in 0,18 Mm CMOS for 2,4 GHz Reader-Less RFID Transponder." Tehnicki Vjesnik - Technical Gazette, Mechanical Engineering Faculty in Slavonski Brod, 2017.

[6] X. Wang, C. C. Boon, K. Yang and L. Kong, "A 20–80 MHz Continuously Tunable Gm-C Low-Pass Filter for Ultra-Low Power WBAN Receiver Front-End," in *IEEE Access*, vol. 9, pp. 154136-154142,

Conclusion

The achieved gain of 400 demonstrates strong amplification capability, while the measured THD of 1.319% confirms acceptable linearity for accurate signal recovery. The total power consumption of 6.261mW reflects efficient for integrated wireless systems. These performance outcomes validate the effectiveness of the proposed 2.45 GHz CMOS receiver architecture in 45 nm technology.

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